

CHIRAL SENSING BASED ON DIRECTIONAL PLASMONS

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During the last decade, the identification of the chiral properties of molecules has grown up to be a hot topic in biosensing [1]. The concept of chirality refers to the lack of specular symmetry in any plane of an object, resulting in the presence of two non-superimposable mirror images or enantiomers. In particular, the interaction between different enantiomers with the human body has deep biological implications, such as the emergence of several health conditions or the appearance of undesired effects with pharmacological compounds, with the latter promoting a growing industrial interest in the topic [2].

Both the investigation of chiral phenomena and the manufacture of chiral drugs require a fine identification of a single molecule chirality or enantiomeric proportion of a sample. This has been typically performed with Circular Dichroism (CD) spectroscopy, measuring the differential absorption of enantiomers when illuminated with left and right-handed Circularly Polarized (CP) light [3]. However, this effect is well-known to be very weak, leading to intensive research in nanophotonics, exploiting nanoantenna effects, to enhance it. This research mainly included approaches based on the use of plasmonic materials, but these have lately been replaced in favour of High Refractive Index Dielectric (HRID) materials due to their low-loss nature and magneto-optic behaviour [4].

Here I will present the results of our most recent research in the topic [5]. By means of FDTD simulation, we demonstrate the capabilities of a mixed metal-dielectric metasurface to offer CD-enhancing capabilities in both the near and far field regimes and to project this enhancement generating Surface Plasmon Polaritons (SPPs) whose direction depends on the incident CP light handedness [6].

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